DA1202H ASSIGNMENT
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ROLL NO:250122019(CST)
DATE:26-03-26

Q1: Create an XML dataset representing a time-dependent process, for example heart-rate monitoring, stock prices, machine sensor readings, rainfall or temperature logs or student attendance trends. Your XML must include noise, extreme spikes, missing timestamps and repeated timestamps. Write R code to:
1. Extract a clean time series
2. Apply at least two outlier detection methods, for example IQR / boxplot rule
3. Compare how many outliers each method detects
4. Visualize normal vs outlier points
5. Justify which method is more appropriate and why.

XML DATASET:(Time dependent process)

```xml
<HeartRateData>
  <Reading><Time>2026-03-01T10:00:00</Time><Value>72</Value></Reading>
  <Reading><Time>2026-03-01T10:01:00</Time><Value>75</Value></Reading>
  <Reading><Time>2026-03-01T10:02:00</Time><Value>300</Value></Reading>
  <Reading><Time>2026-03-01T10:03:00</Time><Value>74</Value></Reading>
  <Reading><Time>2026-03-01T10:03:00</Time><Value>76</Value></Reading>
  <Reading><Time>2026-03-01T10:05:00</Time><Value>73</Value></Reading>
  <Reading><Time>2026-03-01T10:06:00</Time><Value>20</Value></Reading>
  <Reading><Time>2026-03-01T10:07:00</Time><Value>78</Value></Reading>
  <Reading><Time>2026-03-01T10:08:00</Time><Value>77</Value></Reading>
  <Reading><Time>2026-03-01T10:09:00</Time><Value>200</Value></Reading>
</HeartRateData>
```

1) Extracting clean time series(R code)

```r
> data <- xmlParse("heartrate.xml")
> root <- xmlRoot(data)
> time <- xpathSApply(root, "//Time", xmlValue)
> value <- as.numeric(xpathSApply(root, "//Value", xmlValue))
> df <- data.frame(Time = ymd_hms(time), Value = value)
> df <- df %>% arrange(Time)
> df <- df %>%group_by(Time) %>%summarise(Value = mean(Value))
> full_time <- data.frame( Time = seq(min(df$Time), max(df$Time), by = "1 min")
+ )
> df <- full_time %>% left_join(df, by = "Time")
> df$Value <- na.approx(df$Value, na.rm = FALSE)
```

2) Applying two Outlier Detection Methods

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i)IQR method

```
> Q1 <- quantile(df$Value, 0.25)
> Q3 <- quantile(df$Value, 0.75)
> IQR_val <- IQR(df$Value)
> lower <- Q1 - 1.5 * IQR_val
> upper <- Q3 + 1.5 * IQR_val
> df$IQR_Outlier <- df$Value < lower | df$Value > upper
> cat("IQR Outliers:", sum(df$IQR_Outlier), "\n")
IQR Outliers: 3
```

ii) Z-score method

```
> mean_val <- mean(df$Value)
> sd_val <- sd(df$Value)
> df$Z_score <- (df$Value - mean_val) / sd_val
> df$Z_Outlier <- abs(df$Z_score) > 2
> sum(df$Z_Outlier)
[1] 1
```

3) IQR Outliers: 3
   Z-score Outliers: 1

4)Visualization of Normal vs Outliers

```
> ggplot(df, aes(Time, Value)) +
+     geom_line() +
+     geom_point(aes(color = IQR_Outlier), size = 3) +
+     ggtitle("IQR Outlier Detection")
```

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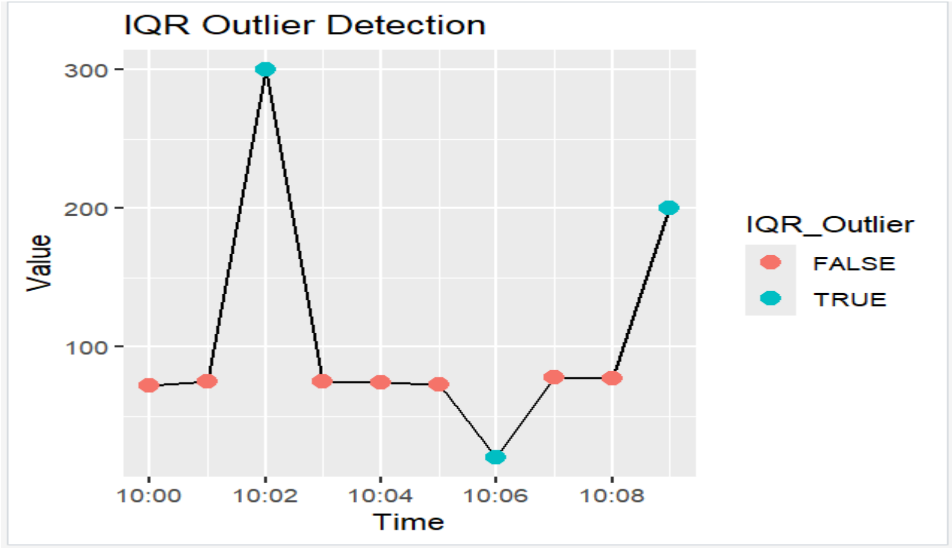
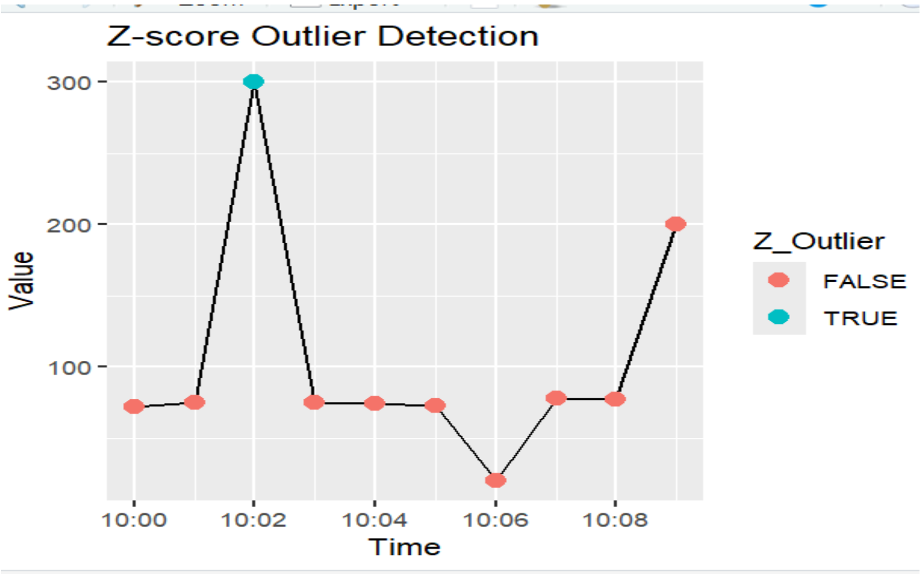



```r
> ggplot(df, aes(Time, Value)) +
+     geom_line() +
+     geom_point(aes(color = Z_Outlier), size = 3) +
+     ggtitle("Z-score Outlier Detection")
```



5).The IQR method is more appropriate for this dataset because it is robust to extreme values and does not assume normal distribution. The dataset contains noise and spikes (e.g., 300 and 200), which distort the mean and standard deviation used in Z-score. Therefore, IQR provides more reliable detection of outliers in real-world time-series data.